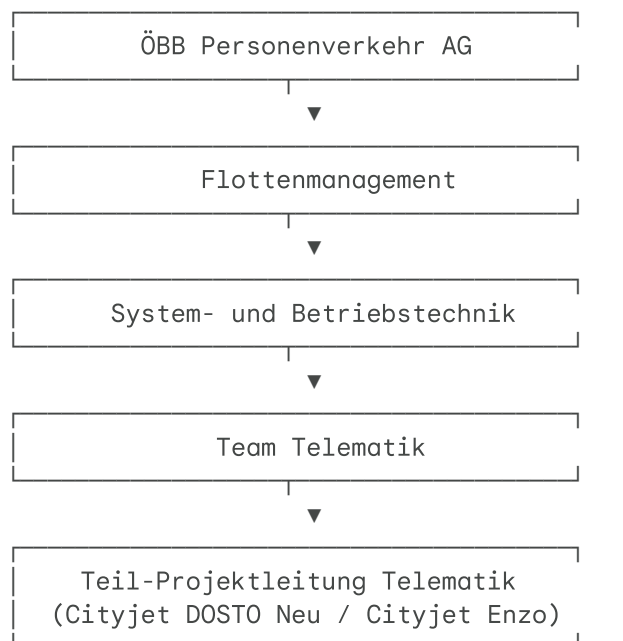


Comprehensive Research Pack: ÖBB Telematics Fleet Management & Automation

1. Persona Deep-Dive: Teil-Projektleitung Telematik

The **Teil-Projektleitung Telematik** (Sub-Project Manager for Telematics) is a highly technical, interdisciplinary coordinator sitting within the **ÖBB-Personenverkehr AG** structure under **Flottenmanagement, System- und Betriebstechnik – Team Telematik**. This role operates directly at the intersection of onboard **Operational Technology (OT)** and landside **Information Technology (IT)**.



Educational and Technical Profile

- **Background:** Advanced degree (HTL, FH, or University) in Telecommunications, Software Engineering, Electrical Engineering, or Network Technology.
- **Domain Expertise:** Highly proficient in Ethernet topologies, routing protocols (e.g., RSTP), wireless link bonding (LTE/5G), virtualization, and edge-computing constraints.
- **Operational Competencies:** Expert in public procurement, writing technical requirements sheets (*Lastenhefte*), Change Request (CR) arbitration, and V-model test structures.

2. Strategic Fleet Profiles

The Sub-Project Manager currently orchestrates two high-visibility, capital-intensive fleet programs:

A. Cityjet Doppelstock Neu (Stadler KISS Platform)

- **Investment Scope:** €1.5 Billion for 109 next-generation double-decker electric multiple units (EMUs).

518 seats). Special 5-car Class 4705 variants are ordered for the City Airport Train (CAT), with 200 km/h-capable long-distance versions (KISS 200) scheduled.

- **Project Stage:** * *Type Approval:* Achieved on April 21, 2026, for Austria, Czech Republic, and Slovakia.
 - *First Deployment:* Maiden passenger journey commenced on May 6, 2026, on the Wien Hauptbahnhof to Wiener Neustadt line.
 - *Sustained Rollout:* 45 regional trains are entering passenger service in the eastern Ostregion (Vienna, Lower Austria, Burgenland) by the end of 2026, primarily running on the high-frequency North-South trunk line (Floridsdorf – Payerbach/Reichenau).
- **Onboard Infrastructure:** Equipped with native ETCS Baseline 3 (v3.6.0), natural refrigerant HVAC heat pumps, gigabit Ethernet backbones, automated passenger counting (APC) cameras, CCTV streams, and passenger info screens.

B. Cityjet Enzo (Fleet Modernization Program)

- **Focus:** Retrofitting and upgrading the existing mid-life commuter fleet (including Desiro ML and Talent EMUs) to achieve system parity with newly delivered trains.
- **Scope:** Injecting modern IoT edge gateways, updating cellular routers to support multi-carrier 5G bonding, deploying AI-enabled APC vision sensors, and integrating legacy subsystems into a unified landside diagnostic data lake.

3. Operational Pain Points & Daily Routines

Primary Pain Points

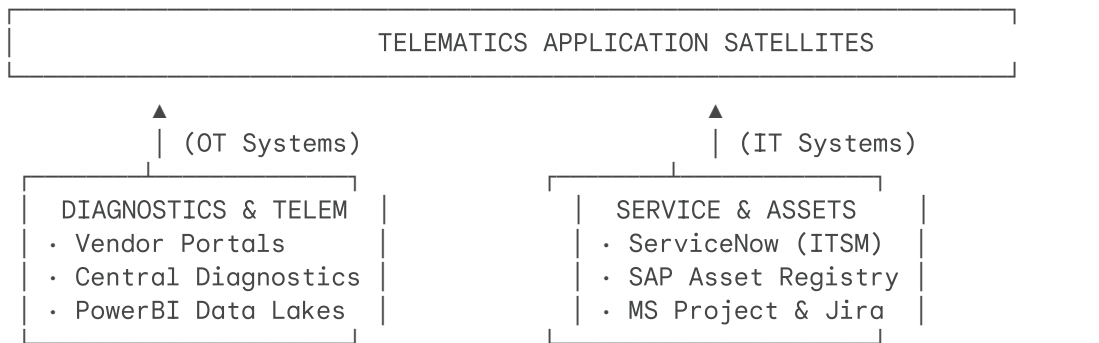
1. **The "Black Box" Vendor Lock-In:** Rolling stock builders (like Stadler or Siemens) protect proprietary bus data. Gaining access to deep diagnostic parameters without triggering massive commercial Change Requests is a constant battle.
2. **Sustained Verification Stress during Trial Runs (*Probetrieb*):** During the initial 2-week active passenger trial phase of a newly delivered train, a single telematics breakdown (e.g., PIS screen mismatch or passenger counting drop) can freeze the formal commissioning process.
3. **Configuration Drift Across the Fleet:** In-service vehicles entering different workshops (e.g., TS-Werkstätte Simmering, Floridsdorf, or Linz) for random repairs often receive localized, inconsistent patch levels.
4. **Connectivity Droop and Environmental Stress:** Trains transit deep alpine tunnels, causing cellular packet loss. Distinguishing between a physical hardware link breakdown and a temporary geographical blackout is difficult without geospatial overlay telemetry.

Refined Daily Routine

- **08:30 – Asset & Coms Check:** Review overnight logs for communication dropouts, version mismatches, and automated edge-fault alerts across both fleets.
- **10:30 – Technical Alignment & Guidelines:** Collaborate with system technicians to draft standardized software configurations and over-the-air (OTA) deployment policies.
- **13:30 – Interdisciplinary Project Board:** Coordinate with procurement teams, overall project leads, and workshop managers to track construction, delivery, and testing milestones.
- **15:30 – Proof of Concept (PoC) Review:** Analyze telemetry data from active test vehicles or digital twin platforms to validate software patches before a fleet-wide rollout.

4. Current Application Landscape at ÖBB

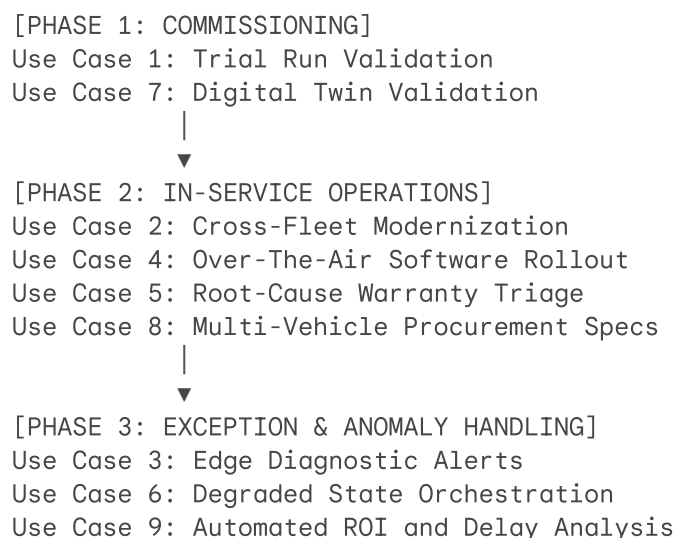
The Telematics Lead's workflow is highly fragmented, requiring them to constantly switch between separate corporate applications:



- **Vendor Diagnostic Portals:** Proprietary tools provided directly by manufacturers (e.g., Stadler's on-train diagnostics or Siemens diagnostic interfaces) to view raw vehicle log files and bus controller parameters.
- **Central Diagnostics Platform (*Diagnoseplattform*):** ÖBB's overarching asset-monitoring portal. Backed by data visualization platforms like Power BI, this tracks high-level communication uptime, passenger counting, and fleet availability.
- **ServiceNow (ITSM):** The centralized service-desk application used to log IT infrastructure tickets, route landside network failures, and assign tasks to back-office engineering groups.
- **SAP Asset & Maintenance (PM/EAM):** The physical maintenance ledger used by **ÖBB Technische Services (ÖBB TS)**. SAP links specific software builds to physical serial numbers, tracking work order completions during scheduled workshop stops (*G-Checks*).
- **Standard Office & Engineering Tools:** Microsoft Project for milestone schedules, Jira/Confluence for software sprint tracking, and Excel for manually reconciling hardware revisions and version matrices.

5. End-to-End System Use Cases

To automate the daily tasks of the Fleet Manager and Telematics Lead, the platform supports 10 distinct operational use cases:



Use Case 1: Commissioning & Trial Run Validation (*Probetrieb*)

- **Context:** Validating a newly delivered train's telematics suite prior to official commercial sign-off.
- **Flow:** The train undergoes a passenger test run. Onboard edge gateways stream continuous telemetry. The platform verifies that passenger info, CCTV, and automated counting systems operate at an error rate of $\text{Err} < 1\%$.
- **Validation:** If a fault is detected, the system displays an end-to-end network path, proving whether the issue is inside an ÖBB landside server or the manufacturer's physical onboard hardware.

Use Case 2: Cross-Fleet Modernization Monitoring

- **Context:** Tracking configuration alignment across retrofitted trains during active operations.
- **Flow:** As trains receive upgrades under the "Enzo" program, the platform automatically logs software versions against physical train IDs (*Triebfahrzeugnummern*), displaying a live compatibility matrix.
- **Validation:** Ensures the team can flag configuration drift before releasing landside updates that could break older hardware revisions.

Use Case 3: Edge Diagnostics & Predictive Alert Management

- **Context:** Intercepting sensor faults before a vehicle enters a maintenance yard.
- **Flow:** Onboard edge platforms run local diagnostic routines (e.g., monitoring door friction curves). If an anomaly is identified, a high-fidelity alert is transmitted to the landside database.
- **Validation:** Allows the team to order targeted maintenance during the vehicle's next scheduled stop, keeping fleet availability high.

Use Case 4: The Over-The-Air (OTA) Software Rollout

- **Context:** Deploying a critical security patch or a passenger information system update across a running fleet.
- **Flow:** The Telematics Lead uses the system to schedule a phased deployment. The update is pushed over cellular channels to a single test vehicle.
- **Validation:** The platform monitors the test vehicle's connection pings for 24 hours. If connection stability remains stable ($P(\text{Uptime}) \geq 99.9\%$), the system safely schedules the rollout across the remaining vehicles in the class.

Use Case 5: Root-Cause Defect Confrontation (Warranty Phase)

- **Context:** Resolving systemic component dropouts under warranty without manual inspection.
- **Flow:** If seat reservation screens on multiple new Stadler double-decker trains go black, the manufacturer may attribute the issue to ÖBB's landside databases.
- **Validation:** The Telematics Lead uses the platform to analyze diagnostic packet logs. This isolates and proves that a memory leak in the manufacturer's physical bus controller is causing the crash, forcing a warranty repair.

Use Case 6: Degraded State Orchestration under Communication Loss

zone or experiences an onboard hardware failure.

- **Flow:** Onboard inference or communication links drop. The landside control board does not show blank screens or trigger false red-alerts.
- **Validation:** The UI enters a graceful, degraded state—holding the last known cached telemetry values, graying out affected nodes, and appending a "Stale Data" indicator badge detailing the elapsed time since the last successful handshake.

Use Case 7: The "Digital Twin" & Cross-Entity Validation (*ÖBB TS Coordination*)

- **Context:** Verifying that a physical hardware component swap in the workshop has not broken software routing.
- **Flow:** Technicians at the depot swap a faulty physical switch in a *Cityjet Enzo* car. They trigger an on-demand test.
- **Validation:** The platform queries the onboard hardware list, comparing it against the vehicle's registered "Digital Twin" schematic. It highlights any configuration discrepancies in the UI, ensuring the train is fully functional before leaving the workshop.

Use Case 8: Standardizing Multi-Vehicle Procurement Specifications

- **Context:** Exporting real-world operational failure logs to build robust requirements for upcoming rolling stock tenders.
- **Flow:** The Telematics Lead reviews recurring communication drops inside alpine tunnels.
- **Validation:** The platform correlates these hardware drops to specific antenna positions on the vehicle roof, exporting a standardized requirements clause (*Lastenheft*) to enforce in upcoming manufacturer bidding packages.

Use Case 9: Automated ROI and Delay-Minutes Analysis

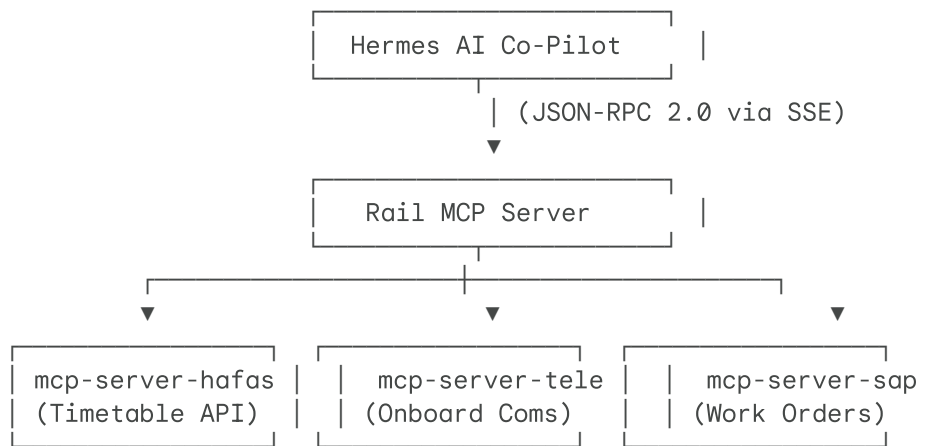
- **Context:** Proving the financial value of the platform to senior stakeholders (such as the Head of Vehicle Maintenance or CFO).
- **Flow:** The system logs every incident where an onboard alert prompted a crew member to resolve an issue before it caused a service delay.
- **Validation:** The platform matches these events against historical delay baselines for identical unmonitored vehicles, calculating and displaying the cumulative **Prevented Delay-Minutes** and total financial savings.

Use Case 10: Closed-Loop Trust Recovery (Damaged Trust Management)

- **Context:** Rebuilding user confidence in the automated alert system after a false alarm.
- **Flow:** If a camera obstruction causes a false "passenger vestibule congestion" alert, the user will naturally trust subsequent alerts less.
- **Validation:** On future alerts of that type, the UI displays a clear accuracy metric (e.g., "*This alert type has an 83% accuracy rating on this route, with 1 false positive in 14 days*"), and provides a high-visibility, neutral **"Dismiss"** option. This transforms manual feedback into an active tuning signal for the system's threshold model.

6. Advanced Automation & Rail MCP Server Architecture

To turn the Hermes AI Co-Pilot into an active operational assistant, the platform exposes a standardized **Model Context Protocol (MCP)** interface. This allows the LLM to interact directly with



Tool Specification (JSON-RPC 2.0 Schemas)

1. Timetable Interface (`mcp-server-hafas`)

Provides the agent with visibility into the scheduled timetable, station stops, and cascading delays.

- `get_train_route_schema` :

```
{
  "name": "get_train_route_schema",
  "description": "Queries real-time scheduled stops, arrival platforms, and de",
  "inputSchema": {
    "type": "object",
    "properties": {
      "train_id": { "type": "string", "description": "e.g., 'TZ-4736-103'" },
      "operating_date": { "type": "string", "format": "date", "description": "
    },
    "required": ["train_id", "operating_date"]
  }
}
```

2. Live Vehicle Telemetry (`mcp-server-fleet-telemetry`)

Allows the agent to query geospatial maps and directly check on-train network parameters.

- `get_fleet_geographic_matrix` :

```
{
  "name": "get_fleet_geographic_matrix",
  "description": "Returns GPS tracks, communication ping latencies, and active",
  "inputSchema": {
    "type": "object",
    "properties": {
      "fleet_type": { "type": "string", "enum": ["DOSTO_NEU", "ENZO"] }
    },
    "required": ["fleet_type"]
  }
}
```

- `trigger_edge_diagnostics :`

```
{
  "name": "trigger_edge_diagnostics",
  "description": "Triggers an on-demand hardware status ping to bypass standar",
  "inputSchema": {
    "type": "object",
    "properties": {
      "train_id": { "type": "string" },
      "subsystem": { "type": "string", "enum": ["COMPRESSOR", "PIS", "EDGE_GAT"] },
    },
    "required": ["train_id", "subsystem"]
  }
}
```

3. Workshop Capacity & Scheduling (`mcp-server-sap-maintenance`)

Connects the agent directly to the maintenance workshops and inventory systems.

- `query_workshop_capacity :`

```
{
  "name": "query_workshop_capacity",
  "description": "Checks available repair slots, staff shift hours, and on-sit",
  "inputSchema": {
    "type": "object",
    "properties": {
      "depot_location": { "type": "string", "enum": ["SIMMERING", "FLORIDSDORF"] },
      "maintenance_type": { "type": "string", "enum": ["TELEMATICS_SWAP", "HAF"] },
    },
    "required": ["depot_location", "maintenance_type"]
  }
}
```

- `reserve_maintenance_slot :`

```
{
  "name": "reserve_maintenance_slot",
  "description": "Creates a temporary, staged work-order reservation in SAP PM",
  "inputSchema": {
    "type": "object",
    "properties": {
      "train_id": { "type": "string" },
      "depot_location": { "type": "string" },
      "arrival_timestamp": { "type": "string", "format": "date-time" }
    },
    "required": ["train_id", "depot_location", "arrival_timestamp"]
  }
}
```

7. The Human-in-the-Loop Safe Action Manifest

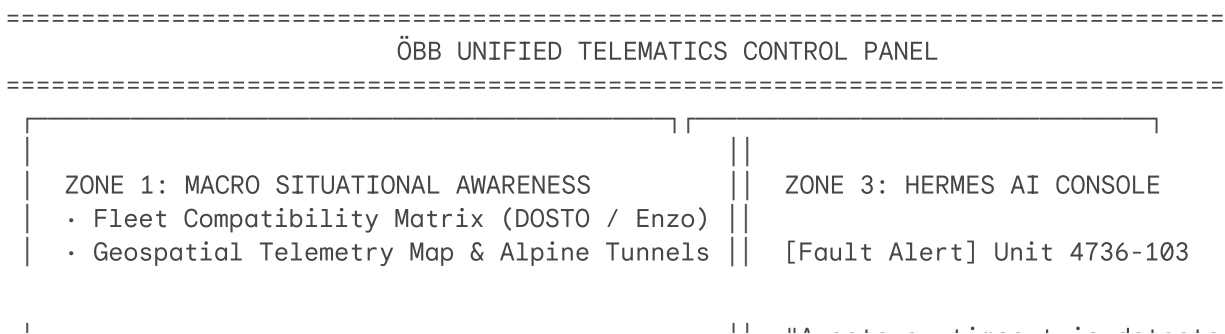
To ensure complete compliance with European railway safety regulations (such as **ECM Directive 2019/779**), the AI agent cannot make changes to databases on its own. Instead, it must bundle multiple tool runs into a single, structured **Action Intent Object**. This object is rendered as an interactive, one-click authorization card in the Fleet Manager's UI.

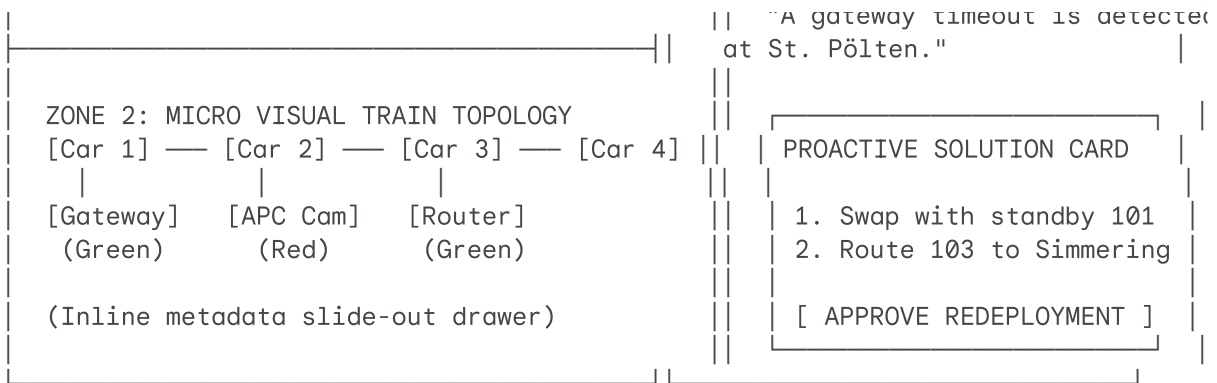
Structured Action Schema

```
{
  "intent_id": "INTENT-20260524-DOSTO-089",
  "action_type": "FLEET_REDEPLOYMENT",
  "justification": "Unit 4736-103 experienced a hard edge-gateway failure at St.
  "operations": [
    {
      "sequence": 1,
      "system": "HAFAS_TIMETABLE",
      "method": "update_rotation_assignment",
      "payload": {
        "retire_unit": "4736-103",
        "inject_unit": "4736-101",
        "location": "WST",
        "execution_time": "22:15:00"
      }
    },
    {
      "system": "SAP_MAINTENANCE",
      "method": "route_to_depot",
      "payload": {
        "unit": "4736-103",
        "destination": "SIMMERING",
        "priority": "HIGH",
        "fault_code": "TEL-EDGE-01"
      }
    }
  ]
}
```

8. UX System & Control Board Architecture

To build a high-performance workspace that minimizes decision fatigue, the UI is organized into three distinct, highly scannable control zones.





Zone 1: Macro Situational Awareness (Fleet Grid & Map)

- **The Interface:** A clean, dense tabular grid tracking the exact network pings, software versions, and telemetry drop rates across the entire *Cityjet Doppelstock Neu* and *Enzo* fleets.
- **Geospatial Layer:** Intermittent telemetry packet drops are automatically plotted onto a geographical map of Austria. If multiple trains drop connection at the exact same location, the UI visually highlights the issue as a regional cellular dead zone rather than a physical vehicle fault.

Zone 2: Micro Visual Train Topology (SVG Anatomy & Drawers)

- **The Interface:** Clicking a vehicle number transitions the workspace into an interactive, block-diagram SVG representation of the physical train carriages (e.g., Cars 1 through 6).
- **Telemetry Node Mapping:** Inside each car, individual hardware components—such as routers, CCTV switches, passenger counters, and PIS monitors—are displayed as color-coded nodes (Green/Amber/Red).
- **Inline Data Drawers:** Clicking a malfunctioning node slides open an analytical drawer, pulling structural metadata directly from SAP: its MAC address, serial number, active firmware version, and the physical switch port it maps to on the train's Ethernet backbone.

Zone 3: Hermes AI Co-Pilot Console (Automation & Logging)

- **The Interface:** A responsive chat and action window integrated with your custom Rail MCP Server, powered by natural language commands.
- **Interactive Solution Cards:** When a critical fault occurs, Hermes parses the variables and displays a pre-compiled **Proactive Solution Card** directly in the sidebar, matching the Action Intent Object format.
- **Human-in-the-Loop Interlock:** The "Approve Redeployment" button on the card serves as a secure interlock. Clicking the button verifies the user's active ÖBB single sign-on (SSO) session, writes an immutable log entry to your central PostgreSQL audit table, and pushes the programmatic commands to the HAFAS and SAP backends simultaneously.

UI Performance & Accessibility Design Targets

1. **Strict Safe-State Latency Target:** Local LLM safety queries running on the train's onboard hardware (via Hailo-10H chips) must hit a performance limit of $P99 < 5$ seconds to ensure real-time responsiveness when transiting areas with poor cellular coverage.
2. **Visual Interaction Sizing for Moving Trains:** Handheld interfaces used by onboard train crew must feature larger tap targets of at least 64 dp (compared to the standard stationary baseline).